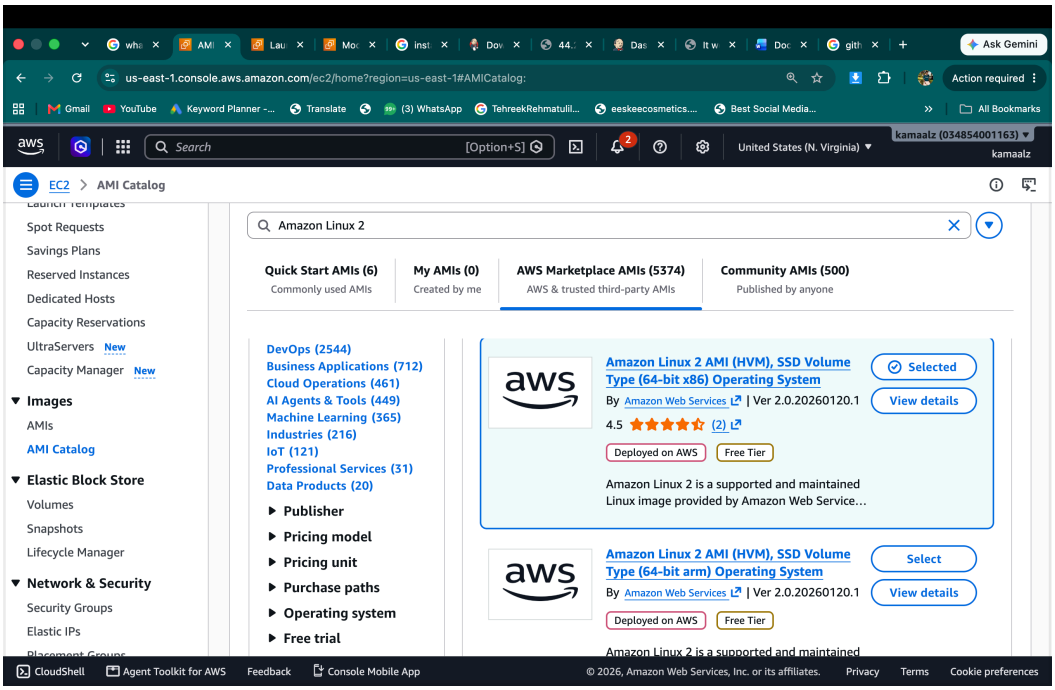
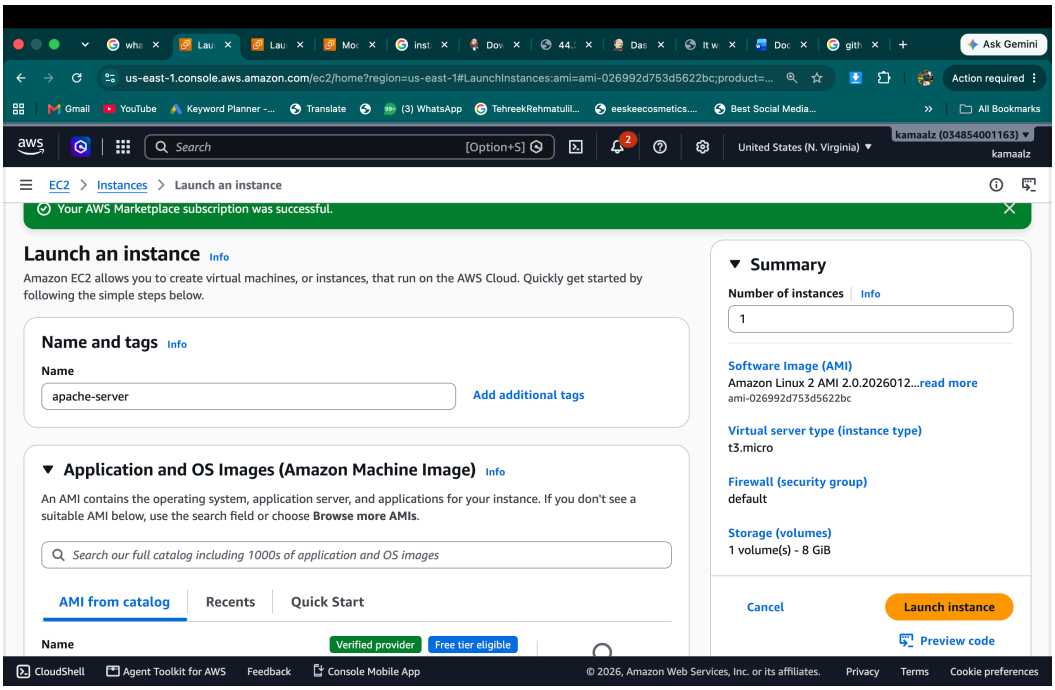
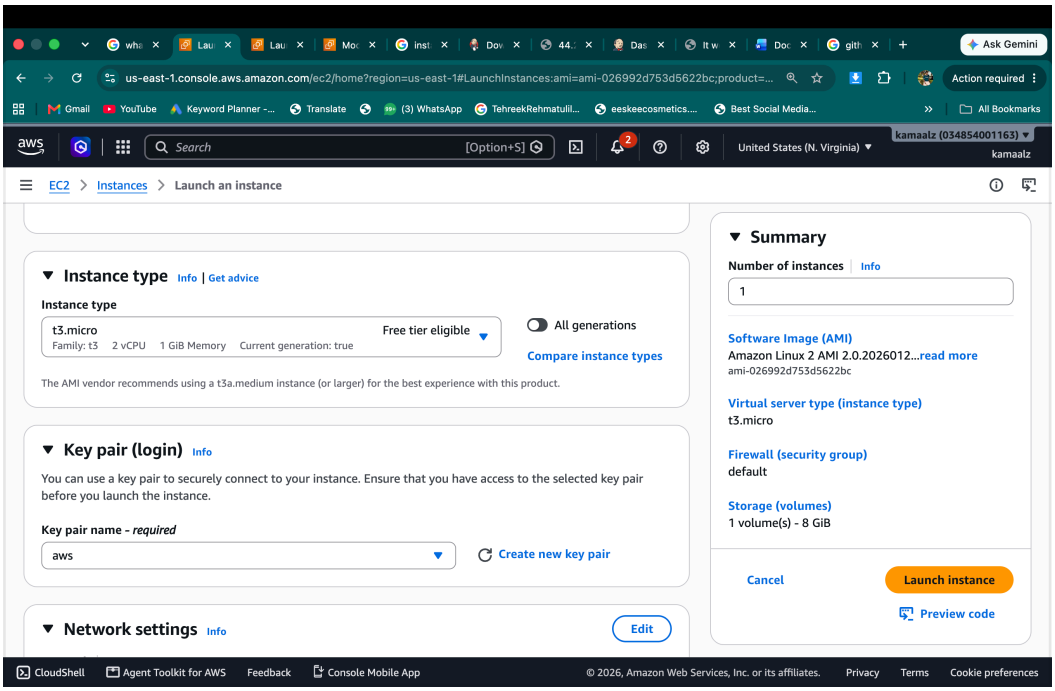
1 Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

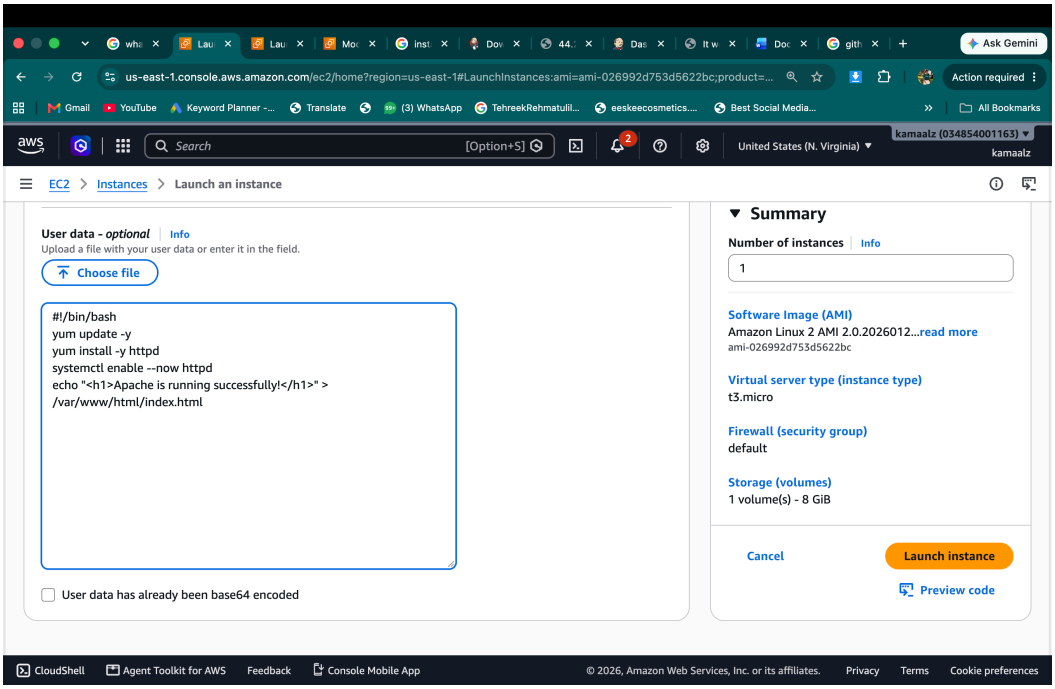
Log in to the AWS Management Console and navigate to EC2 → Launch Instance.

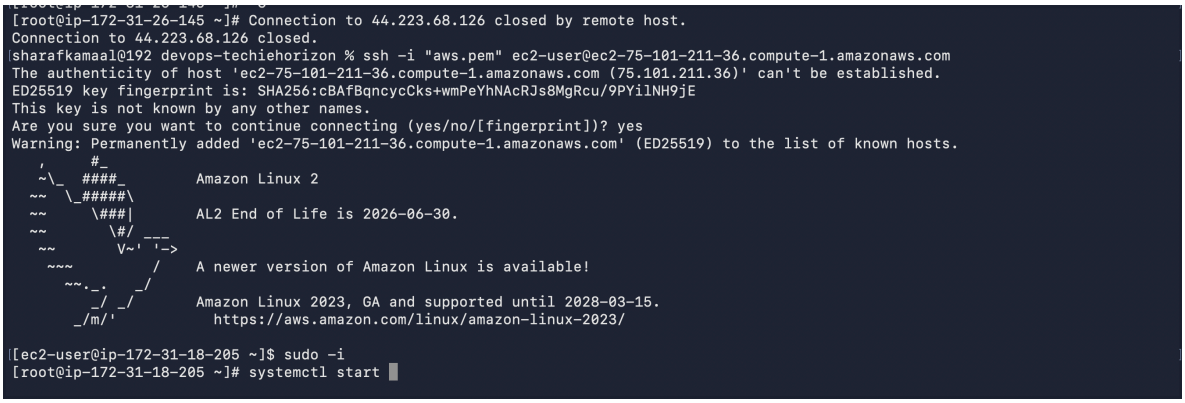
- Enter the instance name as Apache-Server.
- Select Amazon Linux 2 AMI and choose the instance type t2.micro.
- Create or select a Key Pair.
- Configure the Security Group to allow SSH (Port 22) and HTTP (Port 80).
- Expand Advanced Details and enter the following script in the User Data section:
- #!/bin/bash
- yum update -y
- yum install httpd -y
- systemctl start httpd
- systemctl enable httpd
- echo "Apache Server Successfully Installed!" > /var/www/html/index.html
- Click Launch Instance and wait until the instance is running and status checks pass.
- Copy the Public IPv4 address, open it in a browser using http://<Public-IP> and verify that the Apache webpage is displayed.

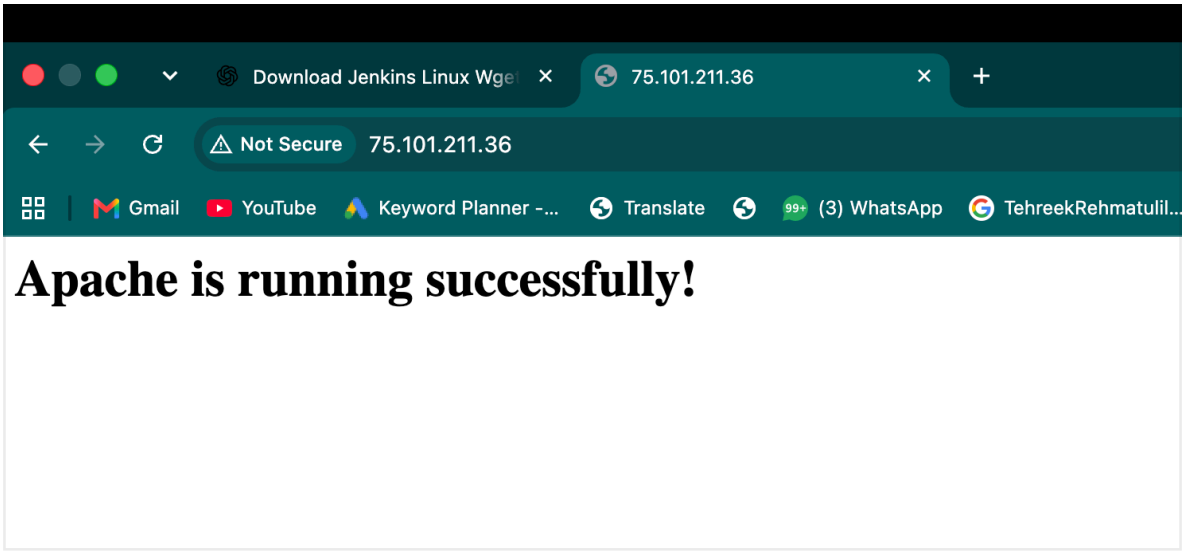



Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:ami=ami-026992d753d5622bc;product=...
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulli...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2
Instances
Launch an instance
Your AWS Marketplace subscription was successful.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
apache-server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
AMI from catalog
Recents
Quick Start
Name
Verified provider
Free tier eligible
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2 AMI 2.0.2026012...read more
ami-026992d753d5622bc
Virtual server type (instance type)
t3.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:ami=ami-026992d753d5622bc;product=...
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulli...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2
Instances
Launch an instance
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
All generations
Compare instance types
The AMI vendor recommends using a t3a.medium instance (or larger) for the best experience with this product.
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
aws
Create new key pair
Network settings Info
Edit
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2 AMI 2.0.2026012...read more
ami-026992d753d5622bc
Virtual server type (instance type)
t3.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



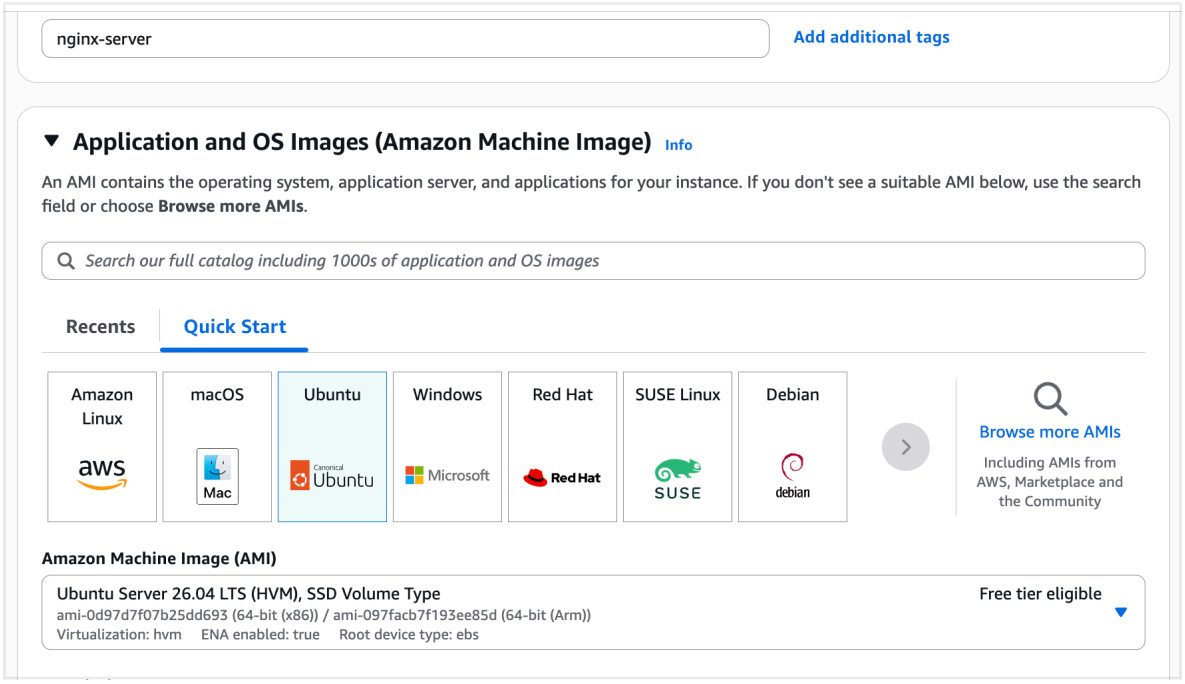




2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

Step 1:- Create a instance. While creating select

Ubuntu image and then
go to advanced option scroll down and we can see user data.
Step 2:- Write the script to install the nginx.
Step 3:- Then after connecting the instance we can check whether the
package is installed or not. We can directly view in etc folder if package is
intsalled or not
3. Launch one Windows server and install Tomcat on Windows.



Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

aws

Create new key pair

Network settings Info

Edit

Network Info

vpc-0ea1bec6073fd383b

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- Create security group
- Select existing security group

Common security groups Info

Select security groups

Compare security group rules

default sg-023167bac6d5470fc
VPC: vpc-0ea1bec6073fd383b

Security groups that you add or remove here will be added to or removed from all your network interfaces.

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit Info

2

Allow tags in metadata Info

Select

User data - *optional* Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash

# Update package lists
apt-get update -y

# Install Nginx
apt-get install -y nginx

# Start Nginx and enable it on boot
systemctl enable --now nginx

# Create a custom test webpage
echo "<h1>Nginx Server is Running Successfully!</h1>" > /var/www/html/index.html
```

User data has already been base64 encoded

```
[root@ip-172-31-18-205 ~]# systemctl start Connection to ec2-75-101-211-36.compute-1.amazonaws.com closed by remote host.
Connection to ec2-75-101-211-36.compute-1.amazonaws.com closed.
sharafkamaal@192 devops-techiehorizon % ssh -i "aws.pem" ubuntu@ec2-54-197-27-130.compute-1.amazonaws.com
The authenticity of host 'ec2-54-197-27-130.compute-1.amazonaws.com (54.197.27.130)' can't be established.
ED25519 key fingerprint is: SHA256:nwyAt4uK+3mFIBbm+Sv5CiktLPBktOVpb+4QK2p/UeE
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-197-27-130.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Oct 28 12:19:38 UTC 2026

  System load:  0.66              Temperature:           -273.1 C
  Usage of /:   30.4% of 6.61GB   Processes:             121
  Memory usage: 27%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.27.185

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

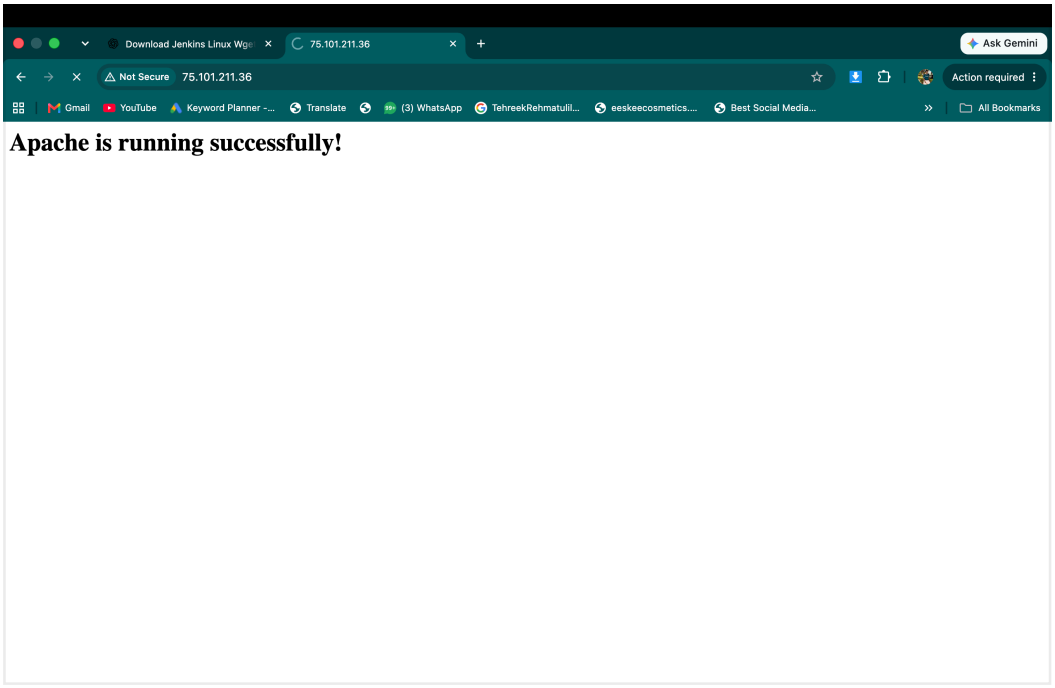
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-27-185:~$
```



3.Launch one Windows server and install Tomcat on Windows.

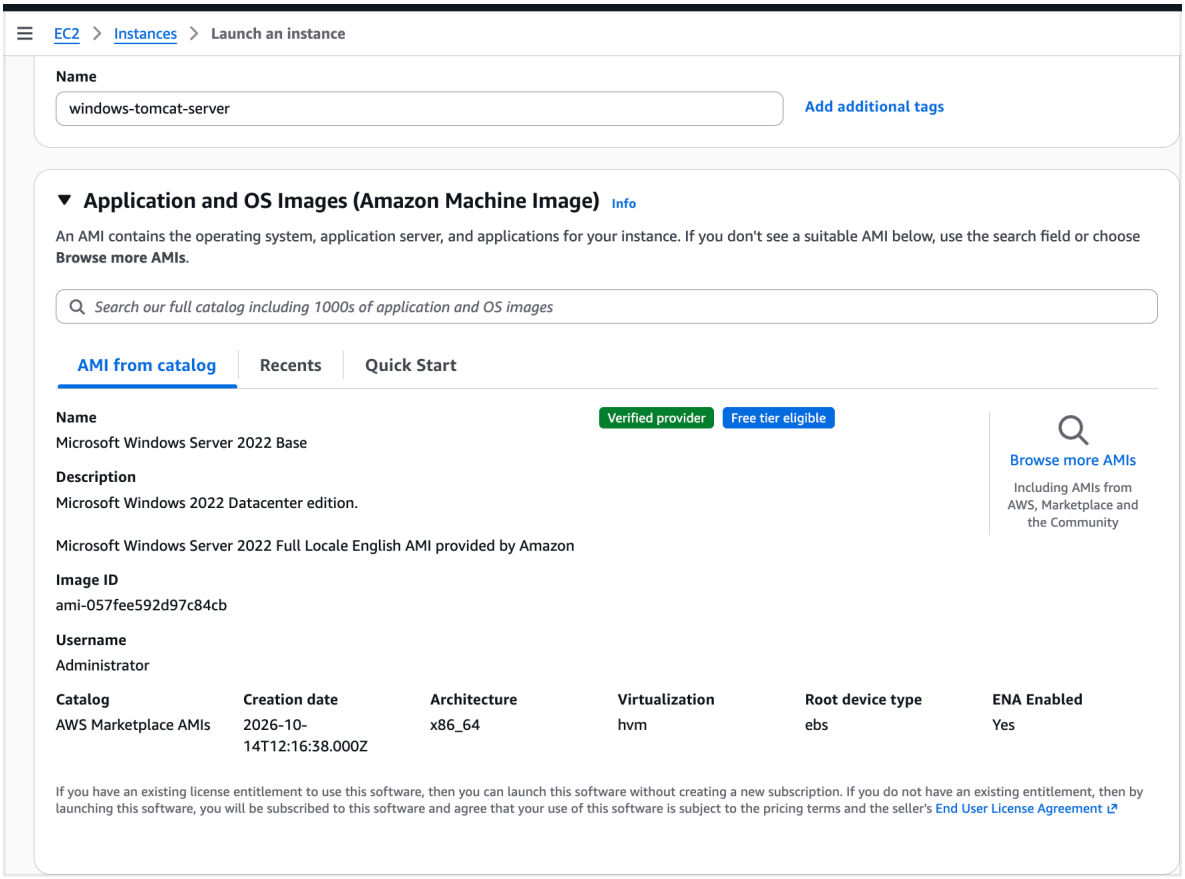
Step 1:- Create a instance with window image. Configure the RDP port in the security group

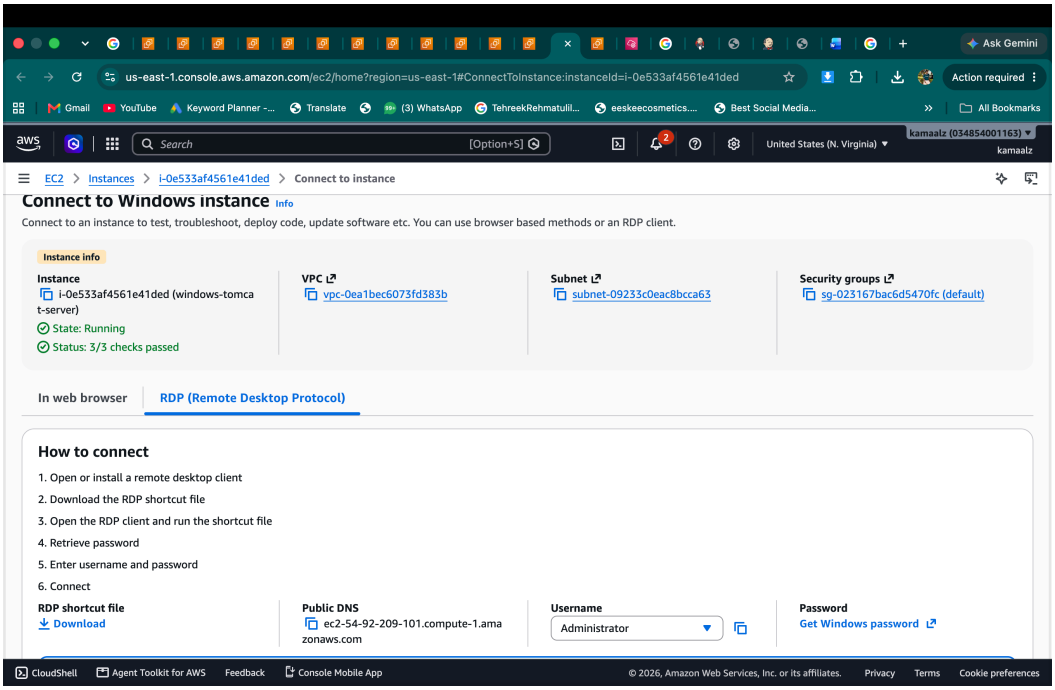
Step 2:- Then using Remote desktop connection connect to the instance.

Step 3:- To install tomcat we need Java installed into the windows

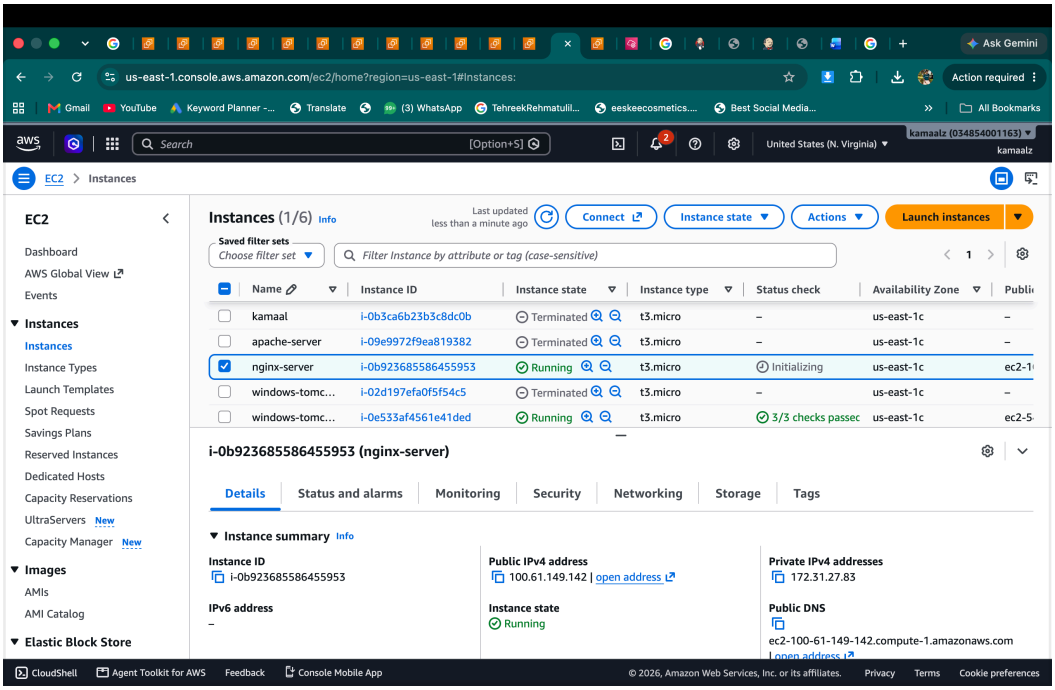
Step 4:- use command prompt and winget to download

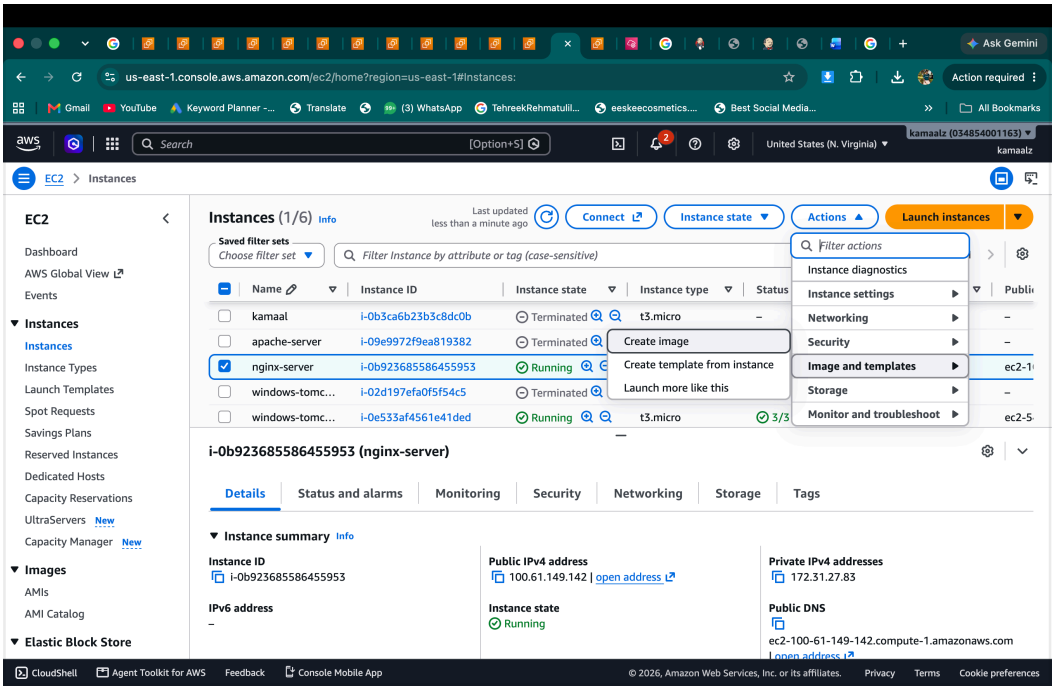
openjdk
Step 5:- Go to web browser and download tomcat installer.
Step 6:- Configure OpenJDK path to tomcat.
Step 7:- After finishing installation we can open localhost:8080 to check
whether it is running or not.

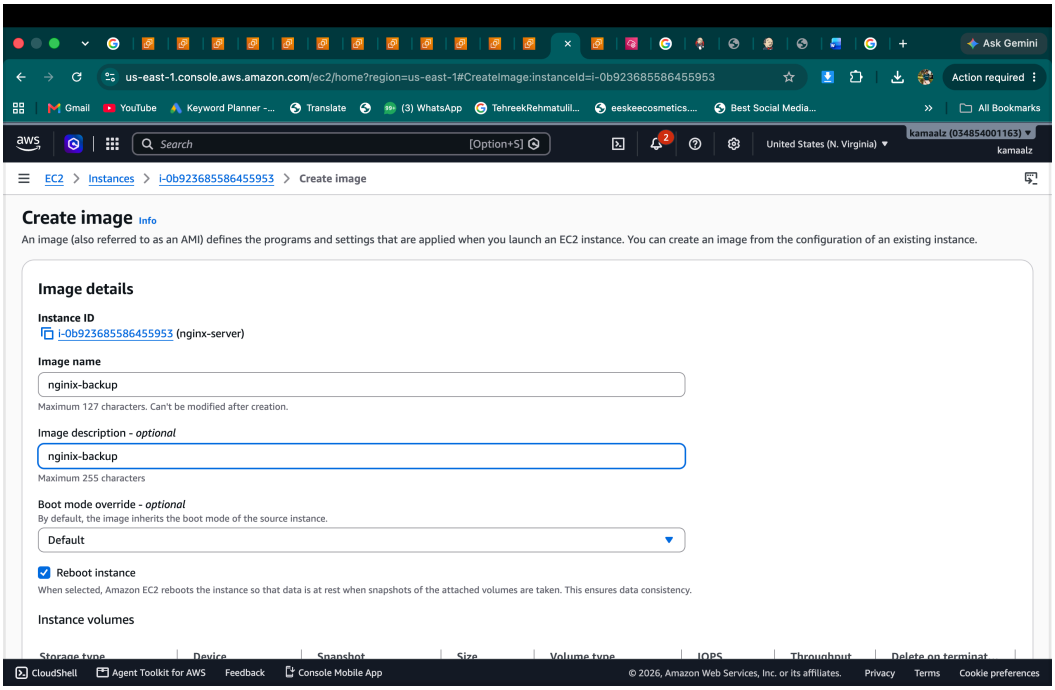


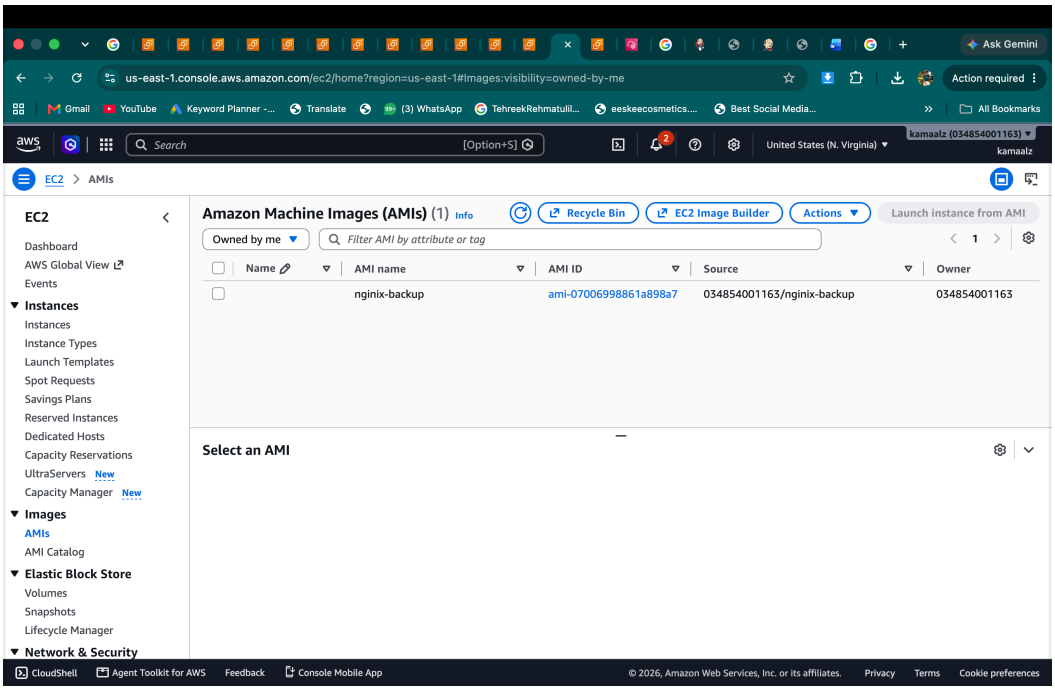


4. Take a snapshot of the instance created in Task 1.
Step 1:- Select the instance go to actions and image
Step 2:- Name the image then scroll down then create image
Step 3:- Now go to AMI to check whether the image is present or not




us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
EC2 > Instances
Instances (1/6) Info
Connect
Instance state
Actions
Launch instances
Create image
Create template from instance
Launch more like this
Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot
i-0b923685586455953 (nginx-server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0b923685586455953
Public IPv4 address
100.61.149.142 | open address
Private IPv4 addresses
172.31.27.83
IPv6 address
Instance state
Running
Public DNS
ec2-100-61-149-142.compute-1.amazonaws.com


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateImage:instanceId=i-0b923685586455953
EC2 > Instances > i-0b923685586455953 > Create image
Create image Info
An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.
Image details
Instance ID
i-0b923685586455953 (nginx-server)
Image name
nginix-backup
Maximum 127 characters. Can't be modified after creation.
Image description - optional
nginix-backup
Maximum 255 characters
Boot mode override - optional
By default, the image inherits the boot mode of the source instance.
Default
Reboot instance
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.
Instance volumes

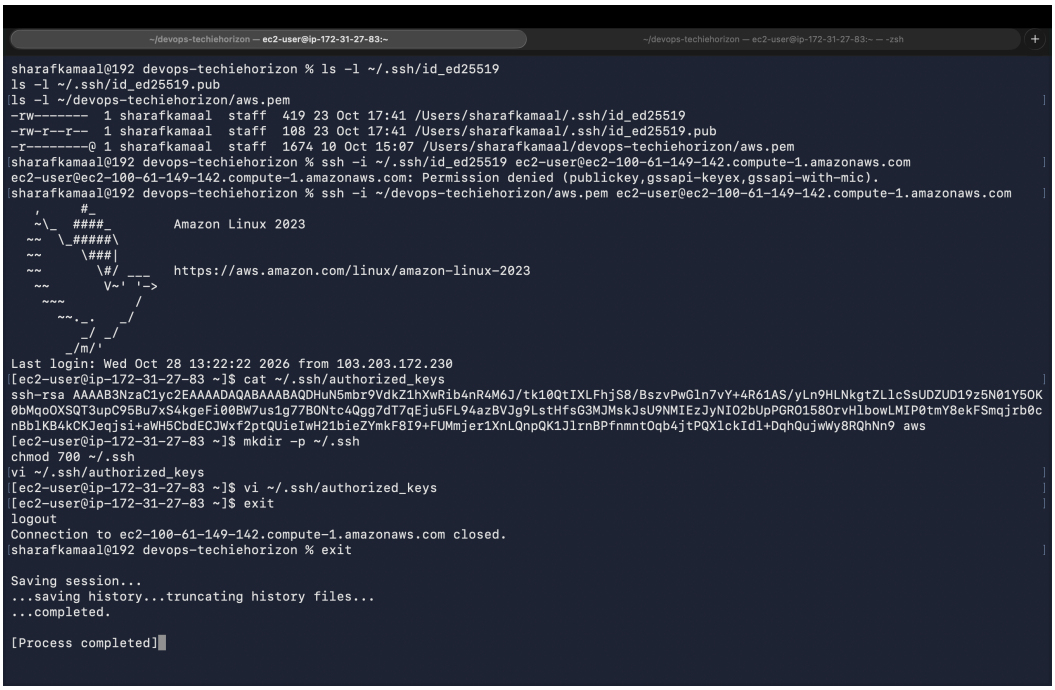


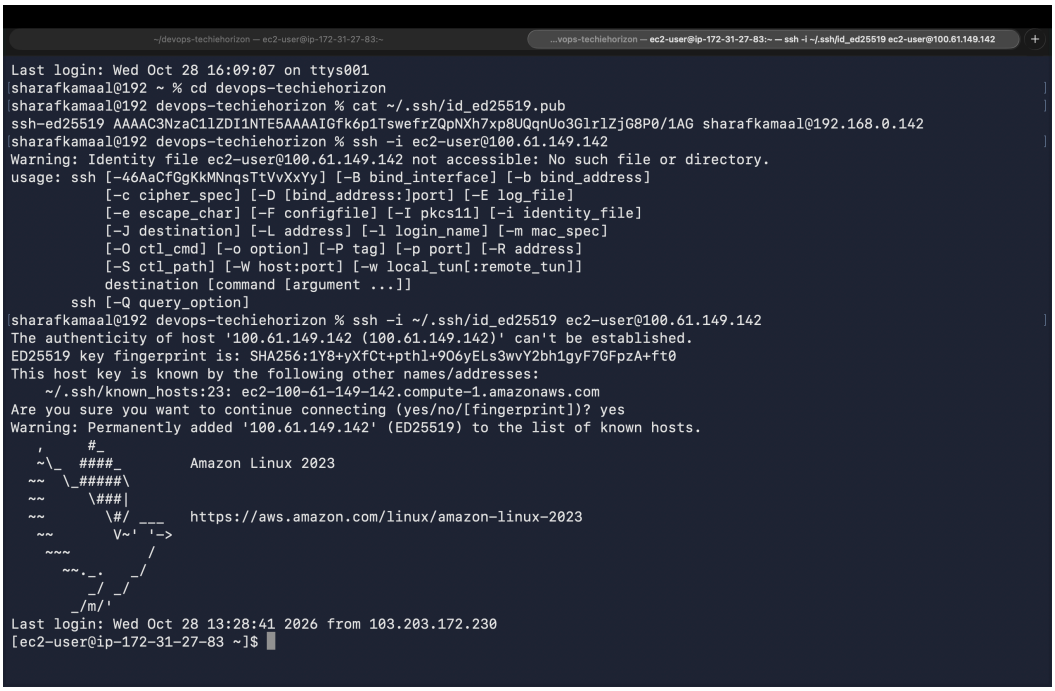
5. Assign passwordless authentication for the EC2 created in Task 2.

Step 1:- First generate a ssh-keygen. Then copy it and paste it into authorized_keys in /root/.ssh/ path

Step 2:- Now go to sshd_config and change PasswordAuthentication from no to yes. Its should not be commented

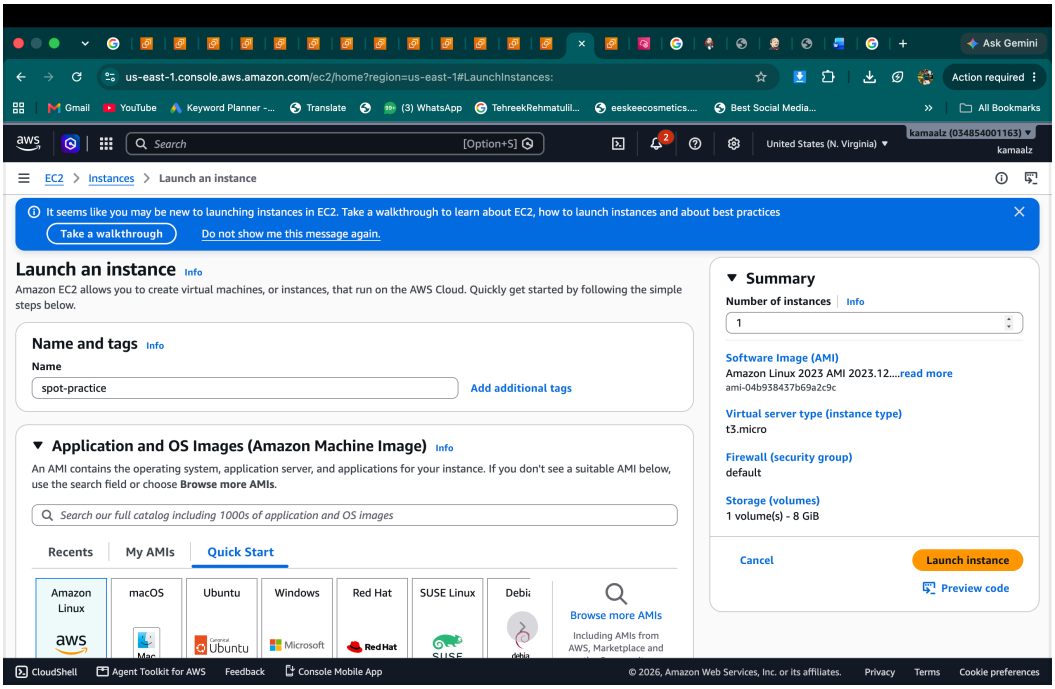
Step 3:- Now try to connect the instance with pem key



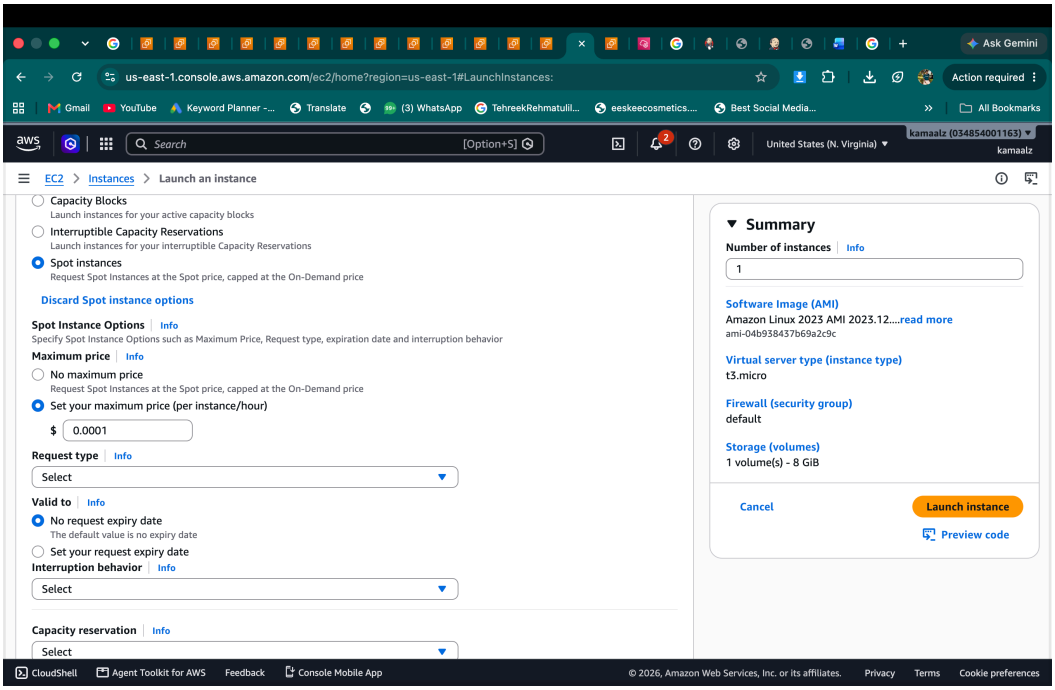


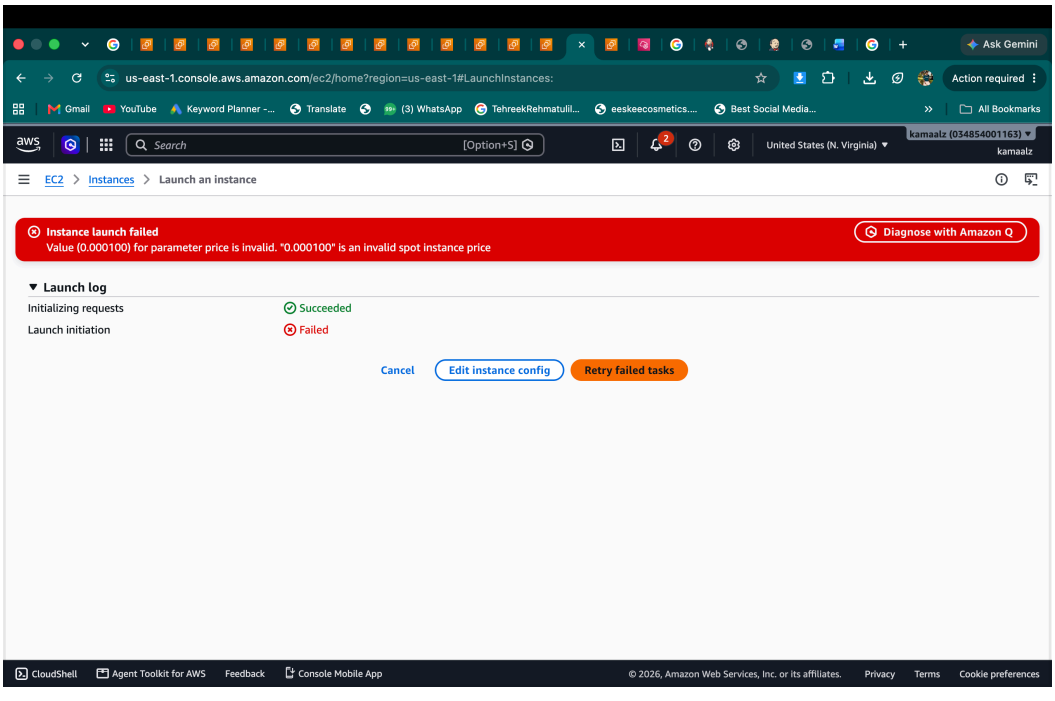
6. Launch any EC2 using the spot purchasing option.

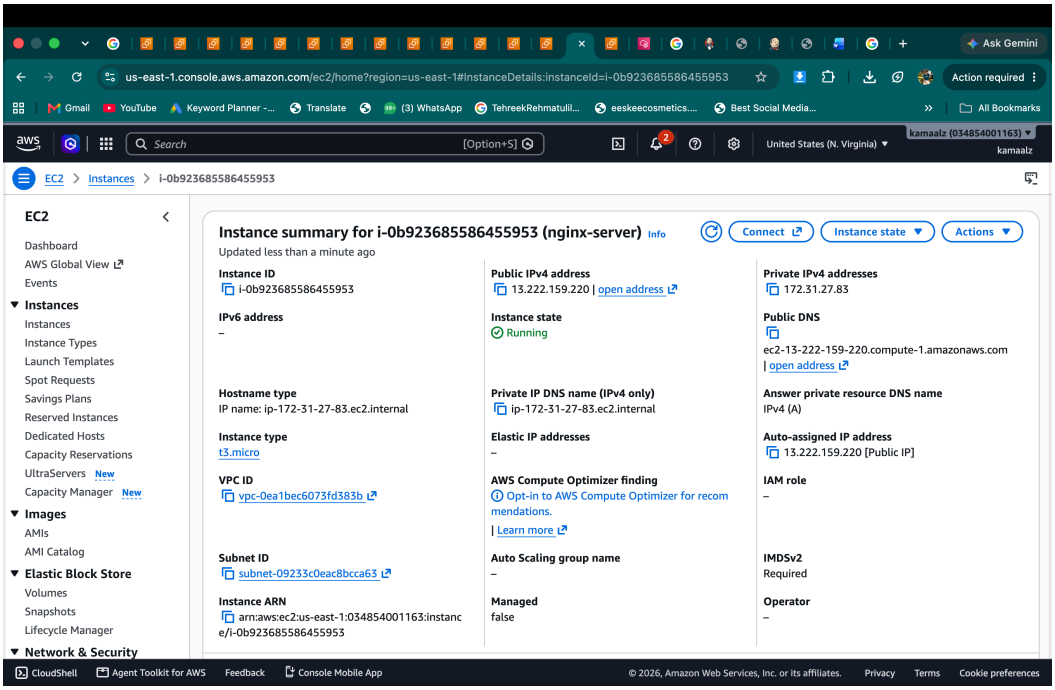
Step 1:- Launch a new instance configure key pair, security groups.
Step 2:- Open Advanced Settings and scroll down to purchase option
select spot instance.
Step 3:- Then launch Instance.


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
spot-practice
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Browse more AMIs
Including AMIs from AWS, Marketplace and
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-04b938437b69a2c9c
Virtual server type (instance type)
t3.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

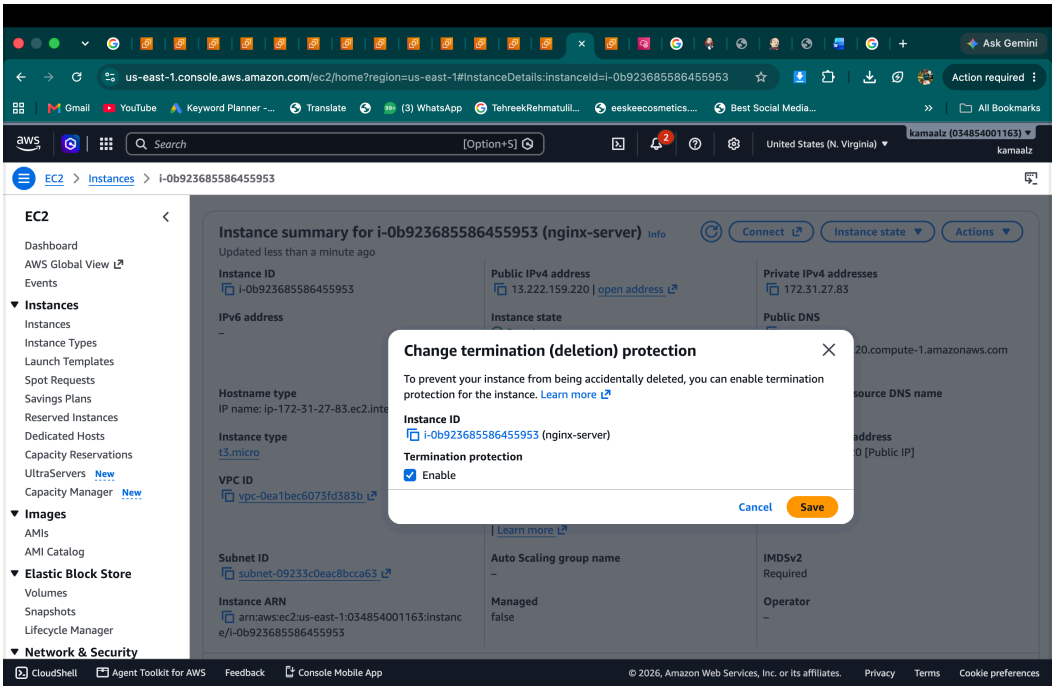



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2 > Instances > Launch an instance
Capacity Blocks
Launch instances for your active capacity blocks
Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations
Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price
Discard Spot instance options
Spot Instance Options | Info
Specify Spot Instance Options such as Maximum Price, Request type, expiration date and interruption behavior
Maximum price | Info
No maximum price
Request Spot Instances at the Spot price, capped at the On-Demand price
Set your maximum price (per instance/hour)
$ 0.0001
Request type | Info
Select
Valid to | Info
No request expiry date
The default value is no expiry date
Set your request expiry date
Interruption behavior | Info
Select
Capacity reservation | Info
Select
Summary
Number of instances | Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-04b938437b69a2c9c
Virtual server type (instance type)
t3.micro
Firewall (security group)
default
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2 > Instances > Launch an instance
Instance launch failed
Value (0.000100) for parameter price is invalid. "0.000100" is an invalid spot instance price
Diagnose with Amazon Q
Launch log
Initializing requests
Succeeded
Launch initiation
Failed
Cancel
Edit instance config
Retry failed tasks
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

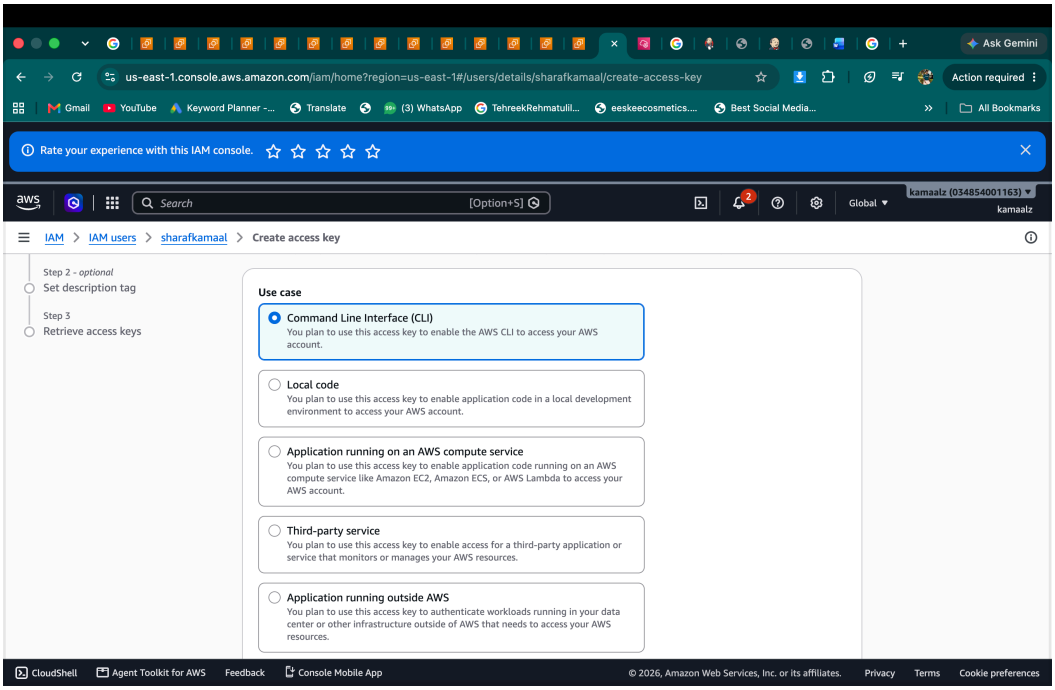


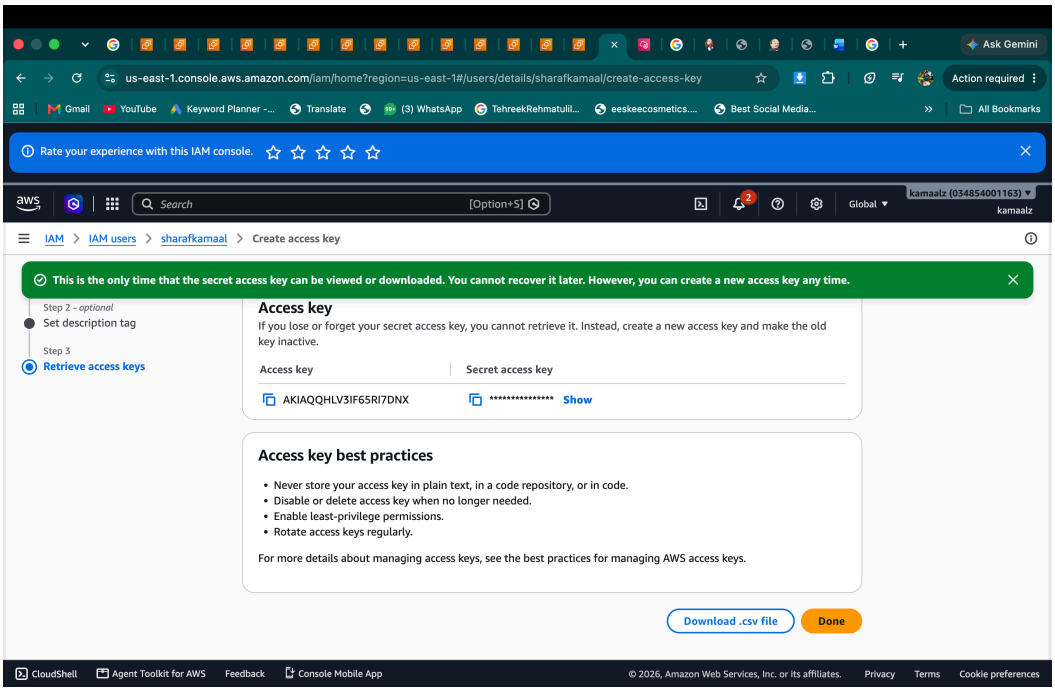
7.Enable termination policy on the EC2 created in Task 2.
Step 1:- Select the instance and then go to action.
Step 2:- In actions go to instance settings and then to change termination protection.
Step 3:- Enable the policy then save.
Step 4:- Now it is successfully enabled

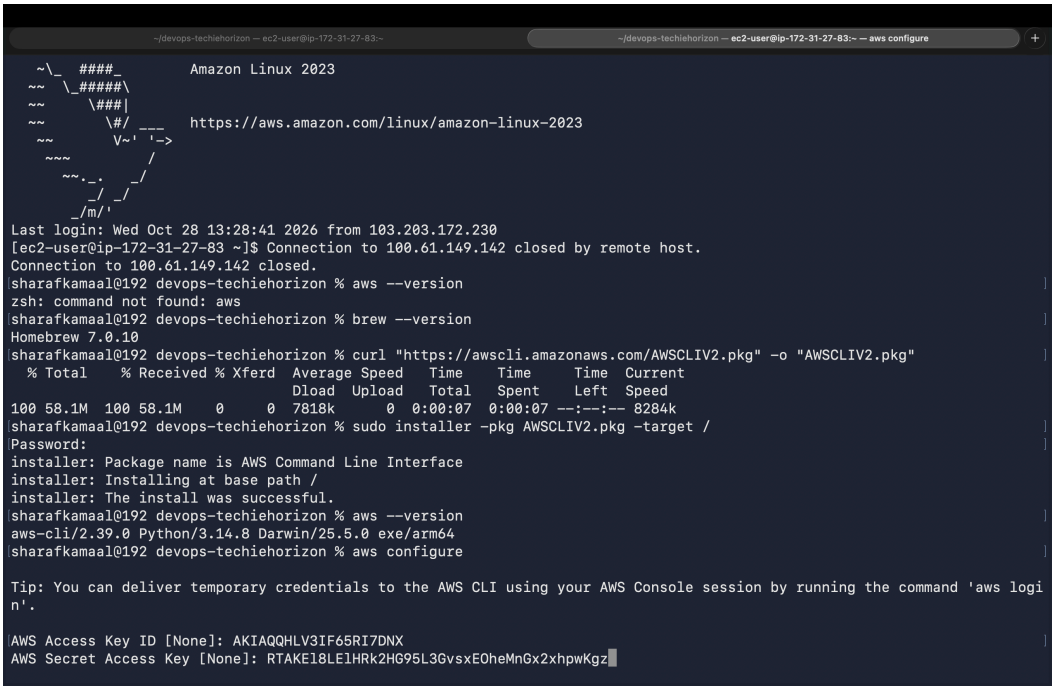


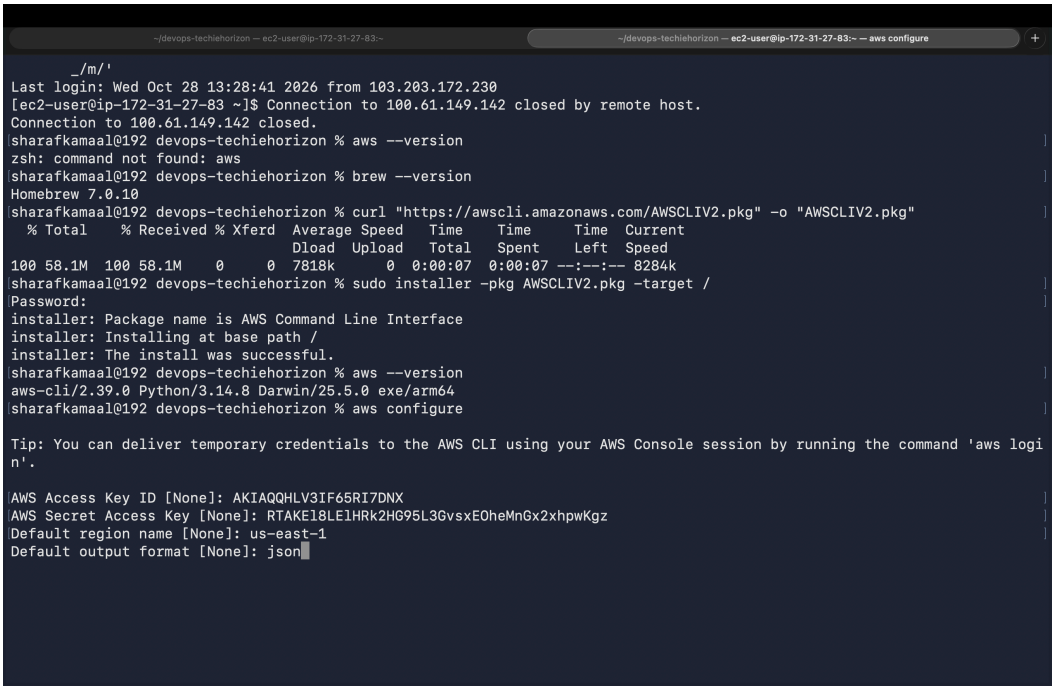
8. Launch one EC2 using AWS CLI.
Step 1:- Create a IAM user with amazonec2fullaccess and admin access
Step 2:-Then click on create access key to create public access key and secret access key.

Step 3:- Open cmd check whether AWS cli is installed or not. If not install it
Step 4:- First configure access key, secret key and availability zone we are using, also output format to json.
Stop 5:- Fetch the AMI image and configure it into run instance command
Stop 6:- Also configure the Pem key file we can select any file.
Stop 7:- We have ssm-parameter and describe image parameter to get AMI id.
Step 8:- After using run instance command instance is created.

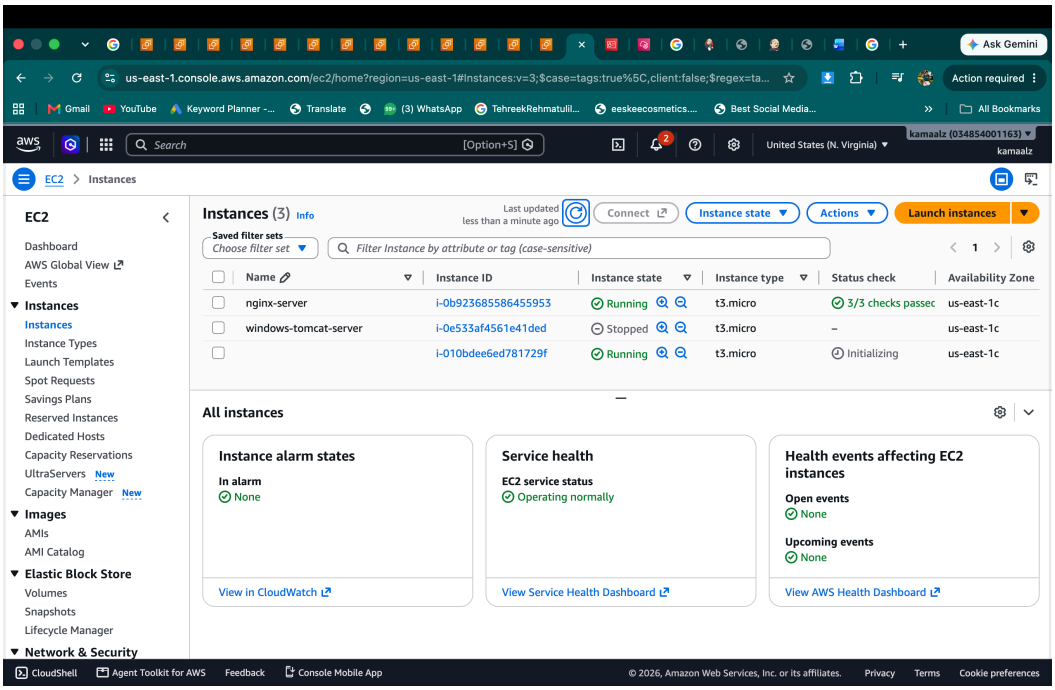









```
aws ec2 run-instances \
 --image-id ami-04b938437b69a2c9c \
 --instance-type t3.micro \
 --region us-east-1
```


Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#instances:v=3;$case=tags:true%5C,client:false;$regex=ta...
Action required
Gmail
YouTube
Keyword Planner -...
Translate
(3) WhatsApp
TehreekRehmatulli...
eeskeecosmetics....
Best Social Media...
All Bookmarks
Search
[Option+S]
United States (N. Virginia)
kamaalz (034854001163)
kamaalz
EC2
Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Instances (3) Info
Last updated
less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
1
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
nginx-server
i-0b923685586455953
Running
t3.micro
3/3 checks passec
us-east-1c
windows-tomcat-server
i-0e533af4561e41ded
Stopped
t3.micro
–
us-east-1c
i-010bdee6ed781729f
Running
t3.micro
Initializing
us-east-1c
All instances
Instance alarm states
In alarm
None
View in CloudWatch
Service health
EC2 service status
Operating normally
View Service Health Dashboard
Health events affecting EC2 instances
Open events
None
Upcoming events
None
View AWS Health Dashboard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Cookie preferences